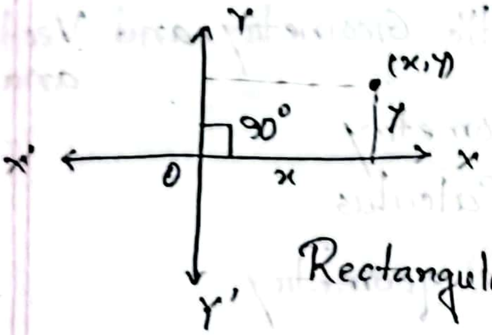
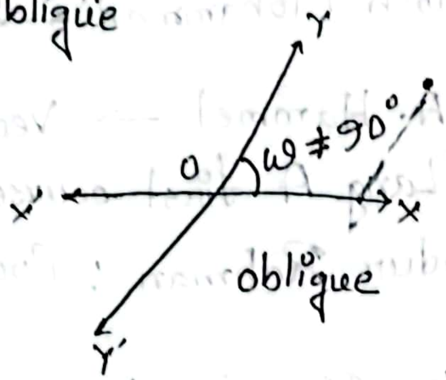
Fermat's last theorem : $x^n + y^n = z^n$ has no integer soln for natural number $n > 2$

Coordinate System :

- Rectangular
- Oblique



Rectangular



(x,y) is not perpendicular to x axis. Only parallel to y axis

oblique

$x \rightarrow$ ~~absesa~~ abscissa

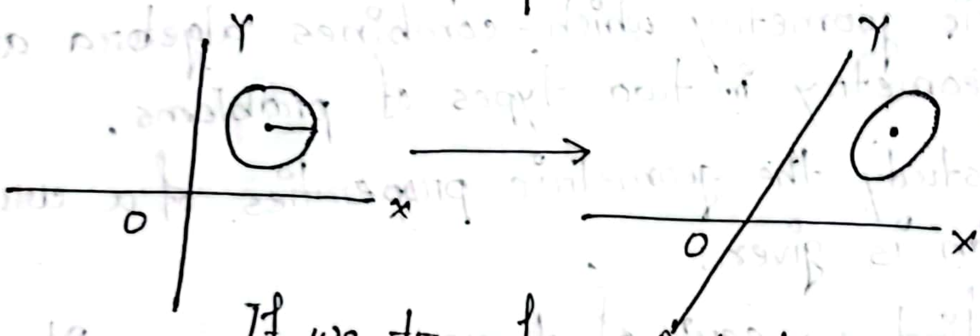
$y \rightarrow$ ordinate

$(x_1, y_1), (x_2, y_2)$

Distance : $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Distance :

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + 2(x_1 - x_2)(y_1 - y_2)\cos\omega}$$

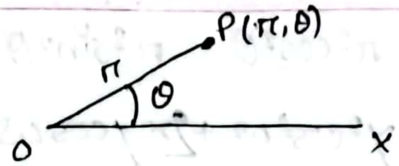


If we transfer a circle to oblique coordinate from a rectangular coordinate it turns to ellipse.

[redacted]

[redacted]

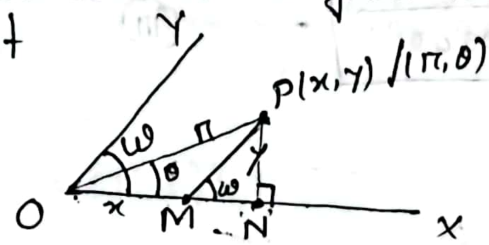
Polar coordinate:



O – Pole

OX – Initial line

Relation between rectangular coordinate and polar coordinate of a point



Let, P is a point on the plane whose coordinate is (x, y) or (r, θ), $\angle XOY = \omega$

$OP = r$

Draw $PM \parallel YO$, $PN \perp OX$

Since, $PM \parallel YO$, $\angle XOY = \angle XMP = \omega$

$OM = x$

$PM = y$

$OP = r$

$\angle XOP = \theta$

and $\angle PMN = \omega$

Now from the right angled triangle

OPN, $ON = OM + MN$

$$\Rightarrow \frac{ON}{OP} \cdot OP = OM + \frac{MN}{PM} \cdot PM$$

$$\Rightarrow r\cos\theta = x + y\cos\omega$$

$$\therefore \boxed{r\cos\theta = x + y\cos\omega} \quad \text{--- (I)}$$

Also from ΔPMN,

$$\frac{PN}{PM} = \sin\omega$$

$$PN = PM\sin\omega$$

$$\Rightarrow \frac{PN}{OP} \cdot OP = y\sin\omega$$

$$\Rightarrow \boxed{r\sin\theta = y\sin\omega} \quad \text{--- (II)}$$

$(i)^2 + (ii)^2 \Rightarrow r^2\cos^2\theta + r^2\sin^2\theta = (x + y\cos\omega)^2 + (y\sin\omega)^2$

$$\Rightarrow r^2 = x^2 + y^2\cos^2\omega + 2xy\cos\omega + y^2\sin^2\omega$$

$$= x^2 + y^2 + 2xy\cos\omega$$

$$\therefore \boxed{r = \sqrt{x^2 + y^2 + 2xy\cos\omega}} \quad --- (iii)$$

$(ii) \div (i)$ করে,

$$\frac{r\sin\theta}{r\cos\theta} = \frac{y\sin\omega}{x + y\cos\omega}$$

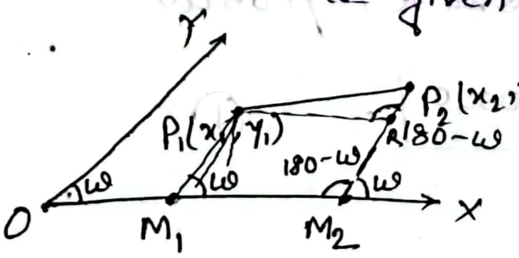
$$\tan\theta = \frac{y\sin\omega}{x + y\cos\omega}$$

$$\therefore \boxed{\theta = \tan^{-1}\frac{y\sin\omega}{x + y\cos\omega}} \quad --- (iv)$$

If, $\omega = 90^\circ$, $(iii) \Longrightarrow r = \sqrt{x^2 + y^2}$

$(iv) \Rightarrow \theta = \tan^{-1}\frac{y}{x}$

Distance between two points in a plane where coordinates ~~acod~~ are given



$P_1(x_1, y_1)$, $P_2(x_2, y_2)$, $\angle XOY = \omega$,

Draw $P_2M_2 \parallel OY$, $P_1M \parallel OY$

$\angle XM_2P_2 = \omega = \angle XM_1P_1$

So, $\angle M_1M_2P_2 = 180 - \omega = \angle P_1RP_2$

$P_1R = M_2M_1 = OM_2 - OM_1 = x_2 - x_1$

$P_2R = P_2M_2 - RM_2 = y_2 - y_1$

and $\angle P_2RP_1 = \angle OM_2P_2 = 180^\circ - \omega$

Now, by trigonometry (cosine rule) we have for the triangle P_1P_2R

$$(P_1P_2)^2 = (P_2R)^2 + (P_1R)^2 + 2P_2R.P_1R\cos(180-\omega)$$

$$= (y_2-y_1)^2 + (x_2-x_1)^2 + 2(x_2-x_1)(y_2-y_1)\cos(180-\omega)$$

$$= (y_2-y_1)^2 + (x_2-x_1)^2 + 2(x_2-x_1)(y_2-y_1)\cos\omega$$

$$\therefore \boxed{P_1P_2 = \sqrt{(y_2-y_1)^2 + (x_2-x_1)^2 + 2(x_2-x_1)(y_2-y_1)\cos\omega}} \quad \text{---} (1)$$

If $\omega = 90^\circ$, (1) $\Rightarrow$ $P_1P_2 = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

• Find the distance between two points $(2,-3)$ and $(-5,-7)$, where the axes being inclined at 60°.

$$d = \sqrt{(2+5)^2 + (-3+7)^2 + 2\cdot(2+5)(-3+7)\cos 60^\circ}$$

$$= \sqrt{49+16+28}$$

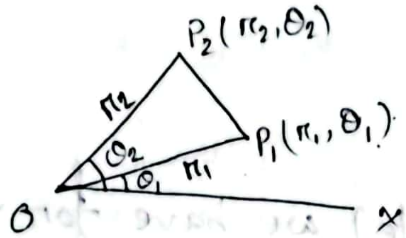
$$= \cancel{\sqrt{37}}\ \sqrt{93}$$

$\cancel{49}$
$\cancel{28}$
$\cancel{21}$
$\cancel{16}$
$\cancel{37}$

49
28
16
93

[redacted] [redacted]

Distance between two points in polar co-ordinates :-



Let, $P_1(r_1, \theta_1)$ and $P_2(r_2, \theta_2)$

$OP_1 = r_1$, $OP_2 = r_2$

$\angle XOP_1 = \theta_1$, $\angle XOP_2 = \theta_2$

$\therefore \angle P_1OP_2 = \theta_2 - \theta_1$

Now for triangle OP_1P_2 ;

$$P_1P_2^2 = OP_1^2 + OP_2^2 - 2 OP_1 \cdot OP_2 \cos(\angle P_1OP_2)$$

$$(P_1P_2)^2 = r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_2 - \theta_1)$$

$$P_1P_2 = \sqrt{r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_2 - \theta_1)}$$

\# Find the distance between two points $(3, 20^\circ)$ and $(5, 80^\circ)$.

The distance between two points P_1 and P_2 is,

$$P_1P_2 = \sqrt{r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_2 - \theta_1)}$$

$$= \sqrt{3^2 + 5^2 - 2 \cdot 3 \cdot 5 \cos(80 - 20)}$$

$$= \sqrt{19} \text{ units.}$$

Hence,

$P_1 = (3, 20^\circ)$

$P_2 = (5, 80^\circ)$

$r_1 = 3$

$r_2 = 5$

$\theta_1 = 20^\circ$

$\theta_2 = 80^\circ$

Transform the expression $r^2 = a^2\cos 2\theta$ into cartesian co-ordinates

We know, $x = r\cos\theta$ and $y = r\sin\theta$

We have, $r^2 = a^2\cos 2\theta$

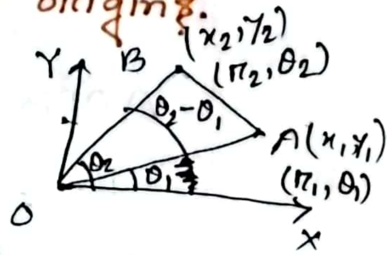
$$r^2 = a^2(\cos^2\theta - \sin^2\theta)$$

$$r^4 = a^2(r^2\cos^2\theta - r^2\sin^2\theta)$$

$$(x^2+y^2)^2 = a^2(x^2-y^2)$$

Area of a triangle whose one vertex is at origin?

Let, OAB is the triangle whose vetices are $O(0,0)$; $A(r_1,\theta_1)$ or (x_1,y_1) , $B(r_2,\theta_2)$ or (x_2,y_2)



Then we have,

$$r_1\cos\theta_1 = x_1 + y_1\cos\omega$$

$$r_2\cos\theta_2 = x_2 + y_2\cos\omega$$

$$r_1\sin\theta_1 = y_1\sin\omega$$

$$r_2\sin\theta_2 = y_2\sin\omega$$

We know,
$r\cos\theta = x + y\cos\omega$
$r\sin\theta = y\sin\omega$

where ω is the angle between the axes.

Now, $\Delta AOB = \frac{1}{2} \times OA.OB\sin(\angle AOB)$

$$= \frac{1}{2} r_1 r_2 \sin(\theta_2 - \theta_1) \quad --- (1)$$

Now, transforming into the cartesian co-ordinate we have :

$$\Delta AOB = \frac{1}{2} r_1 r_2 (\cancel{\cos\theta_2\sin\theta_1}\ \sin\theta_2\cos\theta_1 - \cos\theta_2\sin\theta_1)$$

$$= \frac{1}{2}\left[(r_2\sin\theta_2)(r_1\cos\theta_1) - (r_2\cos\theta_2)(r_1\sin\theta_1)\right]$$

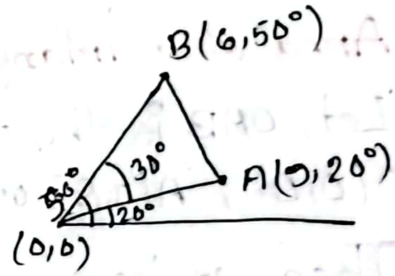
$$= \frac{1}{2}\left[(x_1+y_1\cos\omega)(y_2\sin\omega) - (x_2+y_2\cos\omega)(y_1\sin\omega)\right]$$

$$= \frac{1}{2}\left(x_1 y_2 \sin\omega + y_1 y_2 \sin\omega\cos\omega - x_2 y_1 \sin\omega - y_1 y_2 \sin\omega\cos\omega\right)$$

$$= \frac{1}{2}\sin\omega\left(x_1 y_2 - x_2 y_1\right)$$

$$= \frac{1}{2}\sin\omega\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

Find the area of a triangle whose vertices are (0,0), (9,20°) and (6,50°)

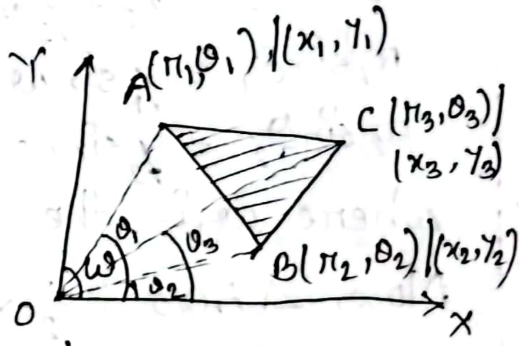


$$\Delta OAB = \frac{1}{2} r_1 r_2 \sin(\theta_2 - \theta_1)$$

$$= \frac{1}{2} \times 9 \times 6 \times \sin 30$$

$= 13.5$ square unit

Area of any triangle :



Area of $\Delta ABC = \Delta OBC + \Delta OAC - \Delta OAB$

$$= \frac{1}{2} r_2 r_3 \sin(\theta_3 - \theta_2) + \frac{1}{2} r_1 r_3 \sin(\theta_1 - \theta_3) - \frac{1}{2} r_1 r_2 \sin(\theta_1 - \theta_2)$$

$$= \frac{1}{2} r_2 r_3 \sin(\theta_3 - \theta_2) + \frac{1}{2} r_1 r_3 \sin(\theta_1 - \theta_3) + \frac{1}{2} r_2 r_1 \sin(\theta_2 - \theta_1)$$

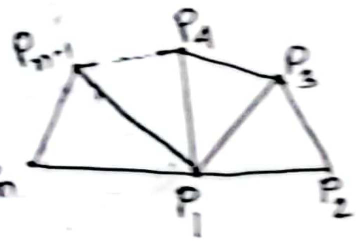
$$= \frac{1}{2}\sin\omega\begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \frac{1}{2}\sin\omega\begin{vmatrix} x_3 & y_3 \\ x_1 & y_1 \end{vmatrix} + \frac{1}{2}\sin\omega\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

$$= \frac{1}{2}\sin\omega\left(x_2 y_3 - x_3 y_2 + x_3 y_1 - x_1 y_3 + x_1 y_2 - x_2 y_1\right)$$

$$= \frac{1}{2}\sin\omega \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

[redacted] **Area of polygon of ~~an~~ sides:** [redacted]

Area of a polygon of n sides whose angular points are $P_1(x_1, y_1)$, $P_2(x_2, y_2)$, ------ $P_{n-1}(x_{n-1}, y_{n-1})$, $P_n(x_n, y_n)$ is $\Delta P_1P_2P_3 + \Delta P_1P_3P_4 + \cdots \Delta P_1P_{n-1}P_n$



$$= \frac{1}{2}\sin\omega\left[(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + \cdots + (x_ny_1 - x_1y_n)\right]$$

Find the area of a polygon whose vertices are $(-5,-2)$, $(-2,5)$, $(2,7)$, $(5,2)$ and $(2,-4)$.

$$\text{Area} = \frac{1}{2}\begin{vmatrix} -5 & -2 \\ -2 & 5 \\ 2 & 7 \\ 5 & 2 \\ 2 & -4 \end{vmatrix} = \frac{1}{2}\left[(-5\cdot5) + (-2\cdot7) + (2\cdot2) + (5\cdot-4) + (2\cdot-2) - (-2\cdot-2) - (2\cdot5) - (5\cdot7) - (2\cdot2)\right]$$

$$= \frac{1}{2}\cdot(-132) = |-56| \text{ sq unit}$$

Ans: 56 sq. unit

If, $\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$ then $\Delta (x_1, y_1), (x_2, y_2)$ and (x_3, y_3) are colinear.

If the points (a, b), (a', b') and $(a-a', b-b')$ are colinear, show that $ab' = a'b$.

$$\frac{1}{2}\begin{vmatrix} a & b & 1 \\ a' & b' & 1 \\ a-a' & b-b' & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{2}\begin{vmatrix} -a' & -b' & 0 \\ a-2a' & b-2b' & 0 \\ a-a' & b-b' & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{2}\left(-a'b + 2a'b' + ab' - 2a'b'\right) = 0$$

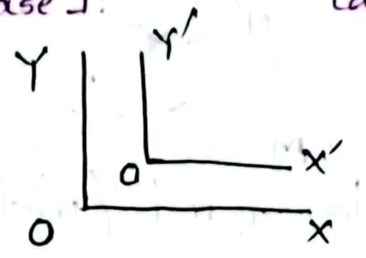
$$\Rightarrow -a'b + ab' = 0$$

$$\Rightarrow ab' = a'b$$

(Shown)

Transformation of coordinates

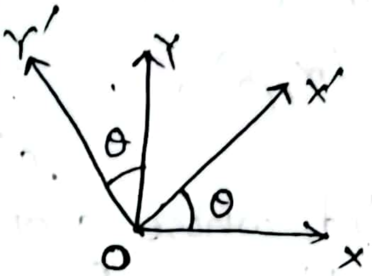
Case 1:



~~Translation~~

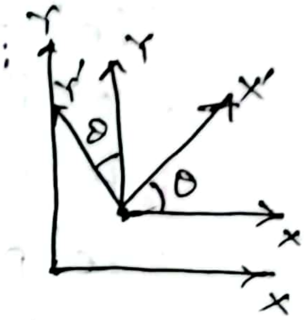
Translation

Case 2:



Rotation

Case 3:



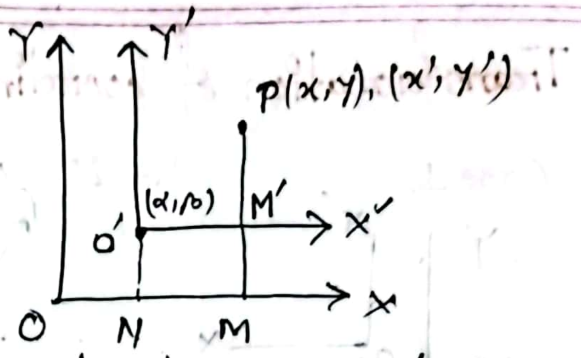
Combination

The coordinate of a point depend upon the origin and the axes of coordinate chosen. The coordinate of a point changes when the origin is changed or the direction of axes changed. The process of changing the coordinate of a point is called the transformation of coordinates.

There are two types of transformation. The first type is known as translation in which the new axes are parallel to the old axes but intersect at a new origin. The second type is known as rotation in which the origin remains same (fixed) but the new axes make a specific angle with the old axes. We will also discuss a case of transformation which is combination of translation and rotation.

i.e → that is

Translation:



Let the origin O' whose coordinate is (α, β).
i.e. $O'(\alpha, \beta)$.

Let P is any point whose coordinate w.r.t (with respect to) old axes is (x, y).

Draw $PM \perp OX$, $O'N \perp OX$

$ON = \alpha$, $O'N = \beta$, $O'M' = x'$, $M'P = y'$

Now, $OM = ON + NM$

$\Rightarrow \quad x = \alpha + O'M' = \alpha + x'$

$PM = MM' + M'P$

$y = NO' + M'P = \beta + y'$

So, $\left.\begin{aligned} x &= \alpha + x' \\ y &= \beta + y' \end{aligned}\right\} \cdots (1)$ or, $\left.\begin{aligned} x' &= x - \alpha \\ y' &= y - \beta \end{aligned}\right\} \cdots (11)$

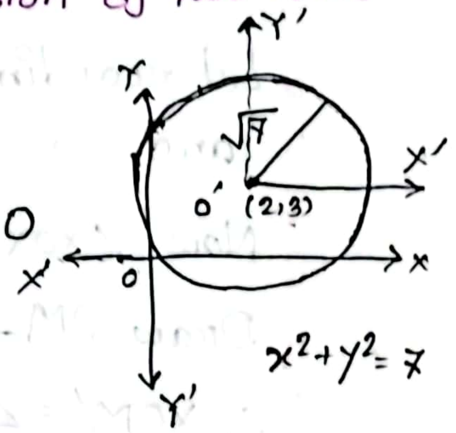
Equ^n (1) and (11) are the equ^n of transformation from old to new and new to old respectively.

[redacted] [redacted]

$x = \alpha + x'$ or $x' = x - \alpha$

$y = \beta + y'$ $\quad$ $y' = y - \beta$

What does the equ^n $x^2 + y^2 - 4x - 6y + 6 = 0$ become when the origin is transfered to $(2, 3)$, The direction of two axes remain same. (unaltered)



$x^2 + y^2 - 4x - 6y + 6 = 0$

$\Rightarrow x^2 - 2x\cdot 2 + 2^2 + y^2 - 2\cdot y\cdot 3 + 9 + 6 - 13 = 0$

$\Rightarrow (x-2)^2 + (y-3)^2 = (\sqrt{7})^2$ ---- (1)

Putting $x = \alpha + x' = 2 + x$ and

$y = \beta + y' = 3 + y$ in equ^n (1) we get,

$(2 + x - 2)^2 + (3 + y - 3)^2 = (\sqrt{7})^2$

$\Rightarrow x^2 + y^2 = (\sqrt{7})^2$

What does the equ^n $x^2 + y^2 - 2x - 2y + 1 = 0$ become when the origin is transferred to $(1,1)$, the direction of axes remain same.

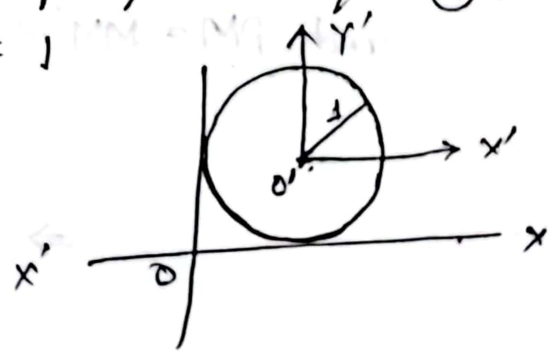
$x^2 + y^2 - 2x - 2y + 1 = 0$

$x^2 - 2\cdot x\cdot 1 + 1 + y^2 - 2\cdot y\cdot 1 + 1 - 1 = 0$

$(x-1)^2 + (y-1)^2 = 1^2$ ---- (1)

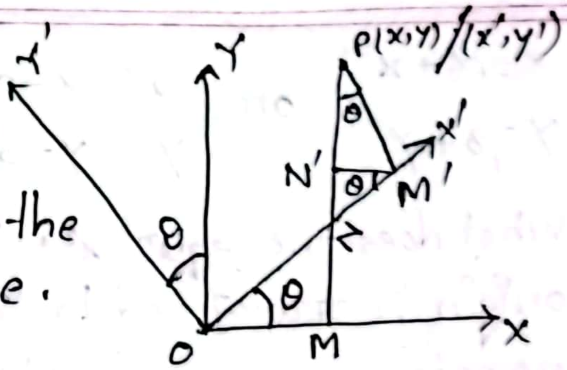
using $x = \alpha + x' = 1 + x$ and $y = \beta + y'$ in equ^n (1) we have, $(1 + x - 1)^2 + (1 + y - 1)^2 = 1$

$\Rightarrow x^2 + y^2 = 1$



Rotation :

Let OX and OY one original axes, they are rotated in the same direction by θ angle.



Let coordinate of P. wrt old axes is (x, y)
and, " of P w.r.t new " " (x', y')

Now $\angle XOX' = \theta$

Draw $PM \perp OX$, $PM' \perp OX'$, $M'N' \perp PM$ $(M'N' \parallel XO)$

$\therefore XOM' = \angle N'M'O = \theta$

and $\angle OM'N' + \angle N'M'P = \pi/2$ --- ①

$\angle N'PM' + \angle PM'N' = \pi/2$ --- ⑪

So from ① and ⑪ we can see,

$\angle OM'N' = \angle N'PM' = \theta$

Now $OM = ON - MN = ON - N'M'$

$$= \frac{ON}{OM'} \cdot OM' - \frac{N'M'}{PM'} \cdot PM'$$

$OM = x$ and $OM' = x'$
$MP = y$. $M'P = y'$

$$\text{So, } x = x'\cos\theta - y'\sin\theta$$

and $PM = MN' + N'P = M'N + N'P$

$$= \frac{M'N}{OM'} \cdot OM' + \frac{N'P}{PM'} \cdot PM'$$

$$\Rightarrow y = x'\sin\theta + y'\cos\theta$$

So, $x = x'\cos\theta - y'\sin\theta$
and $y = x'\sin\theta + y'\cos\theta$ $\Big\}$ --- ①

from

$$x\cos\theta = x'\cos^2\theta - y'\sin\theta\cos\theta$$
$$y\sin\theta = x'\sin^2\theta + y'\sin\theta\cos\theta$$

$$x\cos\theta + y\sin\theta = x'$$
$$x' = x\cos\theta + y\sin\theta$$

Again.

$$x\sin\theta = x'\sin\theta\cos\theta - y'\sin^2\theta$$
$$y\cos\theta = x'\sin\theta\cos\theta + y'\cos^2\theta$$
$(-) \quad (-) \quad (-)$

$$x\sin\theta - y\cos\theta = -y'$$
$$y' = -x\sin\theta + y\cos\theta$$

So, $x' = x\cos\theta + y\sin\theta$
$y' = -x\sin\theta + y\cos\theta$ $\Big\}$ --- ②

	x'	y'
y	$\sin\theta$	$\cos\theta$
x	$\cos\theta$	$-\sin\theta$

$$x = x'\cos\theta - y'\sin\theta$$
$$y = x'\sin\theta + y'\cos\theta$$

and, $x' = x\cos\theta + y\sin\theta$
$y' = -x\sin\theta + y\cos\theta$.

[redacted] [redacted]

Translation and Rotation:

$$x = \alpha + x\cos\theta - y\sin\theta$$
$$y = \beta + x\sin\theta + y\cos\theta$$

Determine the equ^n of the parabolla $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$. If the axes are rotating through $45°$.

When the axes have been rotated through an angle $45°$ then we have,

$$x = x\cos\theta - y\sin\theta = x\cos 45° - y\sin 45°$$
$$= \frac{x}{\sqrt{2}} - \frac{y}{\sqrt{2}} = \frac{1}{\sqrt{2}}(x-y)$$
$$y = x\sin\theta + y\cos\theta = x\sin 45° + y\cos 45°$$
$$= \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = \frac{1}{\sqrt{2}}(x+y)$$

Using the above values in the given equ^n we get,

$$\left(\frac{1}{\sqrt{2}}(x-y)\right)^2 - 2\frac{1}{\sqrt{2}}(x-y)\cdot\frac{1}{\sqrt{2}}(x+y) + \left(\frac{1}{\sqrt{2}}(x+y)\right)^2 + 2\cdot\frac{1}{\sqrt{2}}(x-y) - 4\cdot\frac{1}{\sqrt{2}}(x+y) + 3 = 0$$

$$\Rightarrow \frac{1}{2}(x^2 - 2xy + y^2) - x^2 + y^2 + \frac{1}{2}(x^2 + 2xy + y^2) + \sqrt{2}x - \sqrt{2}y - 2\sqrt{2}x - 2\sqrt{2}y + 3 = 0$$

$$\Rightarrow \cancel{\frac{x^2}{2}} - \cancel{xy} + \frac{y^2}{2} - x^2 + y^2 + \cancel{\frac{x^2}{2}} + \cancel{xy} + \frac{y^2}{2} + \sqrt{2}x - \sqrt{2}y - 2\sqrt{2}x - 2\sqrt{2}y + 3 = 0$$

$$\Rightarrow 2y^2 - \sqrt{2}x - 3\sqrt{2}y + 3 = 0$$

Transform the equn $x^2+3xy+y^2-2x+2y-1=0$ to rectangular axes through the point $(-2,2)$ and inclined at an angle $\pi/4$.

Given the equn $x^2+3xy+y^2-2x+2y-1=0$ ---(I)

We know, $\left.\begin{array}{l} x=\alpha+x\cos\theta-y\sin\theta \\ y=\beta+x\sin\theta+y\cos\theta \end{array}\right\}$ ---(II)

Hence,

$\alpha=-2, \beta=2, \theta=\frac{\pi}{4}$

$x=-2+x\cos\frac{\pi}{4}-y\sin\frac{\pi}{4}=-2+\frac{x}{\sqrt2}-\frac{y}{\sqrt2}=-2+\frac{1}{\sqrt2}(x-y)$

$y=2+x\sin\frac{\pi}{4}+y\cos\frac{\pi}{4}=2+\frac{x}{\sqrt2}+\frac{y}{\sqrt2}=2+\frac{1}{\sqrt2}(x+y)$

Using the above values in equn we have,

$\left[-2+\frac{1}{\sqrt2}(x-y)\right]^2+3.\left[-2+\frac{1}{\sqrt2}(x-y)\right]\left[2+\frac{1}{\sqrt2}(x+y)\right]+$

$\left[2+\frac{1}{\sqrt2}(x+y)\right]^2-2\left[-2+\frac{1}{\sqrt2}(x-y)\right]+2\left[2+\frac{1}{\sqrt2}(x+y)\right]-1=0$

$\Rightarrow \frac{1}{2}x^2+\frac{1}{2}y^2-\sqrt2xy-2\sqrt2x+2\sqrt2y+4-12-3\sqrt2x-3\sqrt2y$

$+3\sqrt2x-3\sqrt2y+\frac{3}{2}x^2-\frac{3}{2}y^2+\frac{1}{2}x^2+\frac{1}{2}y^2+\sqrt2xy$

$+2\sqrt2x+2\sqrt2y+4-\sqrt2x+\sqrt2y+4+\sqrt2x+\sqrt2y+4-1=0$

$\Rightarrow \frac{5}{2}x^2-\frac{1}{2}y^2+3=0$

$\Rightarrow 5x^2-y^2+6=0$